



Cognitive Impacts of Explainable AI in Cybersecurity Incident Response: Challenges and Propositions

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Abstract

Cybersecurity incident response (CSIR) is paramount for organizational resilience. At its core, analysts undertake a cognitively demanding process of data analytics to correlate data points, identify patterns, and synthesize diverse information. Recently, artificial intelligence (AI) based solutions have been utilized to streamline CSIR workflows, notably with an increasing focus on explainable AI (XAI) to ensure transparency. However, XAI also poses challenges, requiring analysts to allocate additional time to process explanations. This study addresses the gap in understanding how AI and its explanations can be seamlessly integrated into CSIR workflows. Employing a multi-method approach, we first interviewed analysts to identify their cognitive challenges, interactions with AI, and expectations from XAI. In a subsequent case study, we investigated the evolution of analysts' needs for AI explanations throughout the investigative process. Our findings yield several key propositions for addressing the cognitive impacts of XAI in CSIR, aiming to enhance cognitive fit to reduce analysts' cognitive load during investigations.

Keywords Cybersecurity analytics · Incident response · Cyber situation awareness · Cognitive overload · Explainable artificial intelligence

1 Introduction

In an era where cyber threats are continuously evolving, cybersecurity incident response (CSIR) has become critical for organizations (Ahmad et al., 2021; H. Naseer et al., 2021). CSIR enables organizations to systematically detect, analyze, and handle security breaches, thus limiting damage and reducing recovery time and costs (Cichonski et al., 2012; IBM, 2023). Due to the constantly changing network environments and cyber threats, CSIR is time-sensitive and requires real-time analytics capabilities (A. Naseer et al., 2021). At the heart of CSIR are cybersecurity analysts (hereafter referred to as analysts), whose responsibilities include triaging the automated detection results (alerts) and compiling information from different data sources to understand

the attacks' tactics, techniques, and procedures (TTPs) (A. Naseer et al., 2021; Zhong et al., 2017). Analysts engage in a cognitively demanding analytical process that includes gathering pertinent data, identifying patterns of incidents, synthesizing information from various sources, and correlating different pieces of data to achieve cyber defense situation awareness (cyber SA) (Erbacher et al., 2010; Kleij et al., 2022; Zhong et al., 2017). The cyber SA enables analysts to gain a comprehensive understanding and project future attack behaviors based on the evolving situation.

Over the years, the CSIR process has increasingly become data-driven, evolving to address the growing complexity of cyber threats (DeCesare, 2022). Modern CSIR tools are now capable of collecting diverse system and device data and aggregating this information for advanced search and analytics. These capabilities enable analysts to correlate data across devices and timeframes, significantly enhancing their ability to detect and respond to complex multi-stage attacks. However, the shift toward data-driven cyber defense strategies has introduced new data challenges. For instance, it is estimated that a 1,000-person organization generates approximately 150 GB of event data daily (Hale, 2012; Martisiute, 2024). This ever-growing data volume

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has resulted in an increasing number of security alerts, with the average organization encountering approximately 5,000 alerts daily (Francis, 2017; Vectra, 2024). Most organizations, however, lack adequately sized teams to handle this alert volume and investigate fewer than half of these alerts (Brozek, 2019; Vectra, 2024). For an average CSIR team of 25 members, this workload translates to approximately 100 alerts per analyst per day. While many of these alerts are ultimately classified as false positives, they still demand time and attention, straining team resources. Over time, the frequent occurrence of false positives, coupled with the sheer volume of alerts, can lead to alert desensitization, where analysts begin to overlook or ignore alerts altogether (Alahmadi et al., 2022). This desensitization increases the organization's overall risk exposure.

Given the increasing volume and complexity of data, a recent effort to enhance the cyber SA capability has been through the application of artificial intelligence (AI) technologies. A primary focus in this area is the development of intelligent incident detection and prevention systems (IDS/IPS). Initially, these systems were built based on traditional machine learning algorithms (Buczak & Guven, 2015), but there has been a significant shift toward employing deep learning techniques for their adaptive learning capability and superior detection accuracy (Gamage & Samarabandu, 2020; Samtani et al., 2023; Sewak et al., 2023). Deep neural networks, in particular, have been applied to tasks such as anomaly detection (Malach et al., 2024; Shoukat et al., 2025), malware detection (Galli et al., 2024; Gulmez et al., 2024), and phishing detection (Biswas et al., 2024), demonstrating remarkable performance in identifying subtle and complex threats. More recently, the research community has been actively investigating the use of Large Language Models (LLM) to further advance cybersecurity professionals' capabilities (Hays & White, 2024; Samtani et al., 2023).

Considering the increasing significance of AI's role in decision-making processes, transparency is critical to building trust and enabling effective collaboration between professionals and AI systems (Merhi, 2023). To address this need, explainable AI (XAI) has gathered significant attention in recent years, aiming to enhance the interpretability and transparency of AI systems (Adadi & Berrada, 2018). XAI addresses the challenges of traditional "black-box" AI models by providing explanations for why AI models make specific predictions or generate certain outcomes (Adadi & Berrada, 2018). These explanations, in turn, improve analysts' understanding and increase their confidence and overall performance (Nyre-Yu et al., 2022). Common XAI techniques, such as LIME (Ribeiro et al., 2016) and SHAP (Lundberg & Lee, 2017), identify the contributions of individual features to a model's predictions. Recognizing its great potential, XAI has been explored in the field of CSIR, serving as a critical tool for enhancing the transparency and

interpretability of AI systems used to detect, analyze, and respond to cyber threats (Gulmez et al., 2024; Hauptman et al., 2023; Holder & Wang, 2021; Liu et al., 2021; Malach et al., 2024; Yayla et al., 2022).

Despite the critical role of analysts in CSIR, minimal research has focused on understanding the cognitive mechanisms underpinning their decision-making (Kleij et al., 2022). The majority of the extant research has been on the system side of CSIR with a focus on incident detection performance (Yayla et al., 2022). However, given the sophistication of cyber attacks, effective data triage requires a compatible human-machine partnership (Albanese et al., 2017). Therefore, understanding how analysts manage large amounts of data and maintain awareness is critical for the design and development of CSIR tools (Paul & Whiteley, 2013). In fact, there have been numerous calls for studies that would increase the performance of cyber SA with the consideration of real-world cyber operations (Albanese et al., 2017; Franke & Brynielsson, 2014; Paul & Whitley, 2013).

XAI introduces an additional layer of complexity to cyber SA. While it enhances the transparency of AI decisions, as discussed earlier, it requires analysts to dedicate additional time to interpreting and processing the explanations generated by XAI systems, referred to hereafter as AI explanations (Holder & Wang, 2021). These new workflow requirements pose significant challenges in the high-pressure and time-sensitive environment of CSIR. We postulate that the demands introduced by XAI may exacerbate the workload of already overburdened analysts, further contributing to existing issues such as burnout and high turnover rates in the cybersecurity field. Recent industry surveys reveal that the average tenure of analysts is decreasing across organizations, with factors such as long shifts, high stress, and alert fatigue cited as key contributors to this trend (Almanza, 2023; Higgins, 2019; Hughes, 2022). This high turnover imposes significant burdens on organizations, as recruiting and training new analysts is both costly and time-consuming. Moreover, these challenges create a self-perpetuating cycle: reduced team capacity places additional strain on remaining analysts, increasing the likelihood of further turnover (Veilberth et al., 2020; Williams, 2022).

Thus, a critical gap remains in understanding how XAI can be effectively integrated into analysts' workflows in a manner that supports decision-making without exacerbating cognitive and operational stress. Addressing this gap is essential not only for improving the efficacy of CSIR processes but also for ensuring the sustainability of the cybersecurity workforce. Recognizing this tension between the need for explanations and its potential detrimental impact, our study aims to dissect the individual workflow of analysts and examine how data challenges evolve with the advent of AI and affect analysts' decision-making.

Building on Information Processing Theory, Cognitive Load Theory, and situation awareness literature, we examined how analysts' cognitive load changes throughout their investigations and how they perceive the role of AI and AI explanations in mitigating cognitive challenges in their workflow. We employed a multi-method approach to address our research questions.

First, we conducted a semi-structured interview study by recruiting eighteen professionals across various industries, all with working experience in incident investigation and response as an analyst. These interviews focused on the cognitive challenges faced during investigations and the participants' expectations and usage of AI technologies. Insights from these interviews highlighted the critical need for explanations of AI-generated results to enhance transparency in the analytical processes. Additionally, they shed light on the varying nature of analysts' information needs due to fluctuating cognitive loads during investigations. However, there remains uncertainty among the professionals regarding the exact impact of these explanations on their investigative workflows. To address this uncertainty and build upon our initial findings, our second case study is designed to explore how analysts' needs for AI explanations evolve throughout the investigative process. We developed an investigation task and recruited six professionals to work on the task while providing them with AI-generated alerts and corresponding explanations. By observing their behavior, especially their interaction with AI explanations, we gained insights into the challenges of integrating AI explanations into the investigative workflow and discovered that this integration is not uniformly advantageous and may introduce unforeseen complications.

Based on our findings from both studies, we have developed key design propositions for addressing the cognitive challenges of XAI in CSIR and to facilitate the effective integration of AI explanations into CSIR workflows. This research contributes both theoretically and practically. Theoretically, our work emphasizes the crucial role of situational context in the effective implementation of AI and automation in cybersecurity. It provides fresh insights into the intricate dynamics of the individual investigation process while interacting with AI systems and AI-generated explanations. Additionally, our study contributes to the XAI literature by discussing the cognitive impacts of explanations and calling for more human-centric designs of XAI tools. It highlights the importance of considering users' goals, experience, and workflows for the successful adoption of these systems. Practically, our findings highlight both the opportunities and pitfalls associated with integrating XAI into the CSIR workflow. The propositions provide a roadmap for more effective and informed design and implementation of XAI in the field.

2 Literature Review and Theoretical Framework

Situation awareness (SA) is a person's mental capacity to understand and draw a conclusion from given environmental elements to make a decision. In his seminal work, Endsley (1988) defined three main phases that reflect progressively increasing SA: *perception* of elements in current situations, *comprehension* of current situations, and *projection* of future status. While SA contributes to effective decision-making, it remains distinct from the decision-making process itself since an individual with high SA may accurately understand their environment yet fail to identify the correct set of actions that are required by the conditions of the environment (Endsley, 1988). Cyber SA can be conceptualized as a subset of SA that requires understanding system activities and the changes from the system baseline (Ofte & Katsikas, 2023; Paul & Whitley, 2013). More specifically, cyber SA involves identifying suspicious indicators, correlating unusual events, and forming hypotheses while considering the tactics, techniques, and procedures employed by attackers to uncover the root cause of an incident (D'Amico & Whitley, 2008).

Analysts employ various tools for intrusion detection, malware detection, access control, and forensics to gather data and information from ongoing system activities. From the situational awareness perspective, the comprehension of the current situation, in other words, the identification of a suspicious indicator, occurs as a function of pattern matching and categorization of available data and information and the existing mental schemas of the analyst. During the decision-making process, analysts continuously update their belief systems through expansion, contradiction, or revision activities (Hettema, 2021). Zhong et al. (2017) proposed a conceptual model named the Action-Observation-Hypothesis Model (AOH Model) to describe the iterative analytical reasoning process of analysts. In this model, analyst actions lead to observations and later to the generation of hypotheses. Subsequently, analysts conduct data triage, aggregate and correlate actions, and gain new evidence, which further updates the hypotheses. As a result of this iterative process, analysts' priorities shift across different stages throughout their analytical process based on the updated information and probabilities. Therefore, effective cyber SA necessitates a fusion of technology and human cognitive abilities. Effective cyber SA, in turn, leads to more rapid incident detection, lowering the impact of a cyber attack (Thangavelu et al., 2021).

From another perspective, the daily routine of an analyst constitutes a series of problem-solving tasks that demand dynamic information processing (Albanese et al., 2017). While analysts can address routine incidents using

memory retrieval type of problem-solving strategies, novel attacks, especially an impasse during decision-making, may require *analytical* (exploitative) or *creative* (exploratory) problem-solving. Incidentally, as sophisticated attacks occur in multiple stages across systems, cyber defense tools generate more data and information for triage, increasing the complexity of the analytical process. The complexities of incident detection and response may in fact require that the updates in analysts' belief systems are done with rationality constraints, as full knowledge of all possibilities and probabilities for a rational decision is often unachievable with the time constraints of the analysis (Hetteima, 2021). Consequently, while cyber SA is an essential part of cyber defense analysis, analysts' cognitive overload is a key hindrance to cyber SA (Hetteima, 2021).

The limited processing capacity of humans is one of the main causes of cognitive overload. Information Processing Theory postulates that the working memory is central to information processing capacity and refers to "the mechanisms and processes that hold the mental representations currently most needed for an ongoing cognitive task available for processing" (Miller, 1956; Oberauer, 2019). The curvilinear inverted U-shaped relationship between available information and receiver performance indicates an optimal amount of information needed for the task at hand for desired outcomes (Driver & Streufert, 1969; Eppler & Mengis, 2004; Hwang & Lin, 1999; Roetzel, 2019). Any information beyond the processing capacity acts like noise for the receiver, leading to distraction, stress, and errors, and consequently, decreases in expected outcomes (Hwang & Lin, 1999; Klapp, 1986). In other words, information overload occurs when decision-makers receive information beyond their processing capacity (Roetzel, 2019). Furthermore, Cognitive Load Theory (CLT) argues that the working memory is not only occupied by intrinsic load (i.e., information that relates to the task at hand) but also occupied by extraneous load (i.e., information that is not relevant to the task at hand, such as distractions) and germane load (i.e., load related to the creation of new mental schemas) (Sweller, 2011).

The application of AI technologies has been extensively researched to reduce the workload on analysts and enhance the efficiency of CSIR (Buczak & Guven, 2015; Gamage & Samarabandu, 2020; Samtani et al., 2023; Sewak et al., 2023). However, the often opaque "black box" nature of these AI systems, combined with the need to adhere to compliance regulations, has propelled the emergence of XAI (Adadi & Berrada, 2018; Biswas et al., 2024; Merhi, 2023). XAI aims to clarify how AI systems generate results and identify the factors considered in making decisions, thus making AI processes more transparent and understandable to the users. Common XAI methods, such as the LIME (Ribeiro et al., 2016) and SHAP (Lundberg & Lee, 2017)

frameworks, explain the contribution of individual features to AI's decisions. Other methods interpret model internals, for instance, by correlating the importance of features at the end of the convolutional layers with the output predictions (Cherepanov et al., 2022).

Table 1 provides a summary of recent advances in the applications of XAI within the cybersecurity domain, such as intrusion detection and malware detection. These studies support the potential of XAI to enhance users' understanding of AI-generated results and the context of these results (Hoffman et al., 2023) and to reduce users' tendency to agree with AI results without verifying whether they are correct (Vasconcelos et al., 2023). However, explanations may not only inadvertently introduce noise and impact users' mental model adjustments in decision-making processes, but they may also lead to confirmation bias depending on whether users agree with the explanations (Bauer et al., 2023). Despite these promising developments, the reviewed literature underscores the need for future research to explore how XAI can be more adaptive to evolving context and user-specific needs. In this regard, XAI-human interaction research, in general and in the context of cybersecurity, as outlined in Table 1, indicates that the impact of XAI on task performance in practice is complex. Therefore, evaluating XAI systems based on the cognitive states of users during the explanation process ensures their effectiveness and practical utility in real-world scenarios (Hsiao et al., 2021). Yet, failing to consider the utility and practicality of explanations for real-world applications is one of the main limitations of the current XAI literature (Abdul et al., 2018).

3 Research Framework and Methodology

Our study focuses on the incident detection and analysis processes performed by analysts in CSIR, who must navigate a multitude of data sources using various cyber defense tools. The dynamic and fast-paced nature of CSIR imposes a significant cognitive load on analysts. In this evolving landscape of data-driven CSIR, it is essential for analysts to employ effective cognitive strategies for task management while also understanding the strengths and limitations of available AI and automation tools. Figure 1 provides a visual overview of the research scope, highlighting the interconnections between data and information, AI-powered cyber defense systems, and analysts. Our investigation builds upon findings from existing literature and industry reports, which emphasize that the increasing volume of machine-generated data from end-user and network devices presents substantial cognitive challenges for analysts, contributing to burnout and turnover in the cybersecurity workforce. While AI-powered systems provide critical support in managing the CSIR workload, they also introduce challenges related

Table 1 Summary of recent advances on the application of XAI in cybersecurity

Literature	Area of Focus	Role of AI	Highlights of XAI
Biswas et al. (2024)	Cyber-risk management	Traditional machine learning techniques (e.g., decision tree) are used to detect phishing attacks	• XAI helps users better understand and assess cyber risks arising from correlated phishing attacks • Simplify multi-class classification problems into two-class scenarios to reduce complexity and improve model explainability • Principal Component Analysis (PCA) aids in identifying significant predictors
Malach et al. (2024)	Anomaly detection	An LSTM model is developed to detect anomalous event sequences	• XAI enhances the interpretability of alerts generated by anomaly detectors • Explanations are provided by visualizing anomalous event sequences as human-interpretable subgraphs and highlighting the most important components
Galli et al. (2024)	Behavioral malware detection	Deep sequential neural networks (i.e., Long-Short Term Memory and Gated Recurrent Unit models) have been applied, leveraging the behavior of malware	• XAI supports the analyses of malware's behavior, thus helping analysts to understand the pivotal malware operations • Example explanations include ranking API calls based on their importance as determined by the model in classifying software behavior • Explanations were provided by XAI techniques, including LIME, SHAP, Layer-wise Relevance Propagation (LRP), and attention mechanisms • XAI lacks qualitative metrics to evaluate the effectiveness of its methodologies directly
Shoukat et al. (2025)	Intrusion detection	An LSTM-based model is used to predict malicious events	• XAI helps analysts understand why specific events are flagged as malicious by the AI model • SHAP techniques are used to provide explanations of the model, thus promoting trust between the stakeholders
Kozik et al. (2024)	Adversarial attacks	A BERT-based method is used to detect fake news	• SHAP is used to provide explanations of the importance of words and phrases that influence model predictions • Adversaries can utilize SHAP explanations by altering the most influential words and phrases to manipulate the model's predictions
Gulmez et al. (2024)	Dynamic malware detection, ransomware detection	Convolutional Neural Network (CNN) architecture is developed to detect ransomware	• XAI highlights the importance of the features in making predictions • LIME and SHAP techniques are used to provide local and global explanations for detection, thus enabling security analysts to identify important indicators of ransomware
Merhi (2023)	Responsible AI	AI has been used in Autonomous and semi-autonomous systems	• XAI enables users to easily understand where the data comes from, the logic behind the model, and the reasoning behind the outputs • Transparency and explainability are interconnected
Vössing et al. (2022)	Human-AI collaboration	AI agents work with humans in decision making	• AI agent transparency reduces the information asymmetry between humans and AI, thus enhancing the trustful and accurate collaboration between humans and AI systems
Hauptman et al. (2023)	Autonomous agents in cyber incident response	AI agents that perform tasks in different stages of cyber incident response	• Work cycles should determine the ideal levels of autonomy and guide an AI agent's changing level of autonomy • Interviews indicate a preference for more explainable AI at higher levels of autonomy • Future research should explore the evolving explainability needs of adaptive autonomous agents

Table 1 (continued)

Literature	Area of Focus	Role of AI	Highlights of XAI
Caro-Martínez et al. (2024)	Personalized experiences of XAI	AI used in decision making and detection scenarios	- One challenge of XAI is to assess the suitability of explanations by monitoring users in real-world work environments - Research is needed to develop user-centered XAI approaches, emphasizing enhanced interactivity and contextual sensitivity to better align with user expectations and needs
Saeed and Omlin (2023)	Challenges of XAI	AI used in critical and sensitive domains (e.g., healthcare and cybersecurity)	- What to explain should align with the user's experience and expertise - Humans must be involved in the creation and use of XAI models - Explanations should be tailored to contexts, taking into account the application domain, the users, the types of explanations, and the most effective methods for delivering them

to explainability, trust, and transparency (Adadi & Berrada, 2018; Ribeiro et al., 2016). Central to our research is XAI and how XAI impacts the cognitive load of analysts.

Existing studies reveal a paradoxical impact of explanations on decision-making. For instance, while explanations can increase decision accuracy, they also significantly prolong decision-making time—in some cases, up to tenfold (Nyre-Yu et al., 2022). Conversely, alerts that lack clear explanations compel analysts to invest additional effort in deciphering their meaning, further extending decision-making time (Alahmadi et al., 2022). Thus, although AI offers promising solutions for handling the growing volume of data in CSIR and XAI addresses the need for AI explainability, there remains a critical gap in understanding how to design and implement XAI systems that support analysts' decision-making without increasing their cognitive load.

More specifically, we aim to address the following research questions:

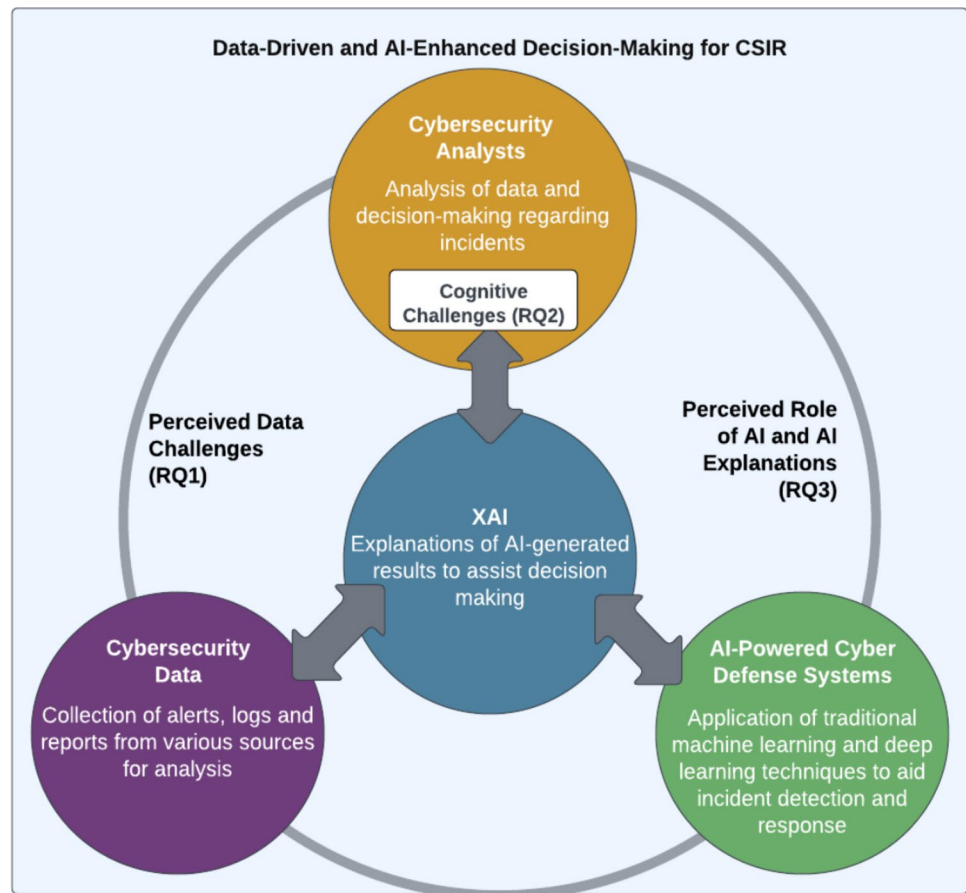
- **RQ1:** How has the role of data and associated data challenges in CSIR evolved over the years from the perspective of cybersecurity analysts?
- **RQ2:** What are the cognitive challenges in the decision-making process of analysts during CSIR?
- **RQ3:** How do cybersecurity professionals perceive the current role of AI and AI explanations in addressing cognitive challenges during CSIR?

To effectively explore these research questions, our research methodology is designed around two primary studies. The first study engages with professional cybersecurity experts through semi-structured interviews, lasting between one to one and a half hours. The focus of the interviews is to determine the impact of the increasing volume of data in the cybersecurity incident investigation context while also exploring the perceptions of professionals regarding the role of AI in mitigating cognitive challenges during data analysis. Following the initial study, a second case study is conducted with an emphasis on exploring how analysts' needs for AI explanations evolve during the investigation process. In this phase, participants are presented with a cybersecurity incident investigation task designed to mimic real-world scenarios. As part of the investigations, we observed their analysis operations as they engaged with the task and paid particular attention to their requests for AI explanations and analyzed the final report of their findings.

4 Study 1: An Interview Study

4.1 Design

Our first study is designed in a semi-structured in-depth interview format, which ensures both the exploration of

Fig. 1 The research framework

predetermined questions and the flexibility for participants to elaborate beyond these queries. Our target demographic comprises individuals employed full-time in cybersecurity, specifically in roles related to data analysis and incident detection and response. To qualify participants, we administer a screening survey that gathers detailed information about their job roles, years of experience, team dynamics, and company background. The interview of select participants is scheduled to last between one hour and one and a half hours, conducted via Zoom to accommodate participants' geographical diversity and schedules. Upon obtaining the participants' consent, we proceed to record the interview sessions for comprehensive analysis.

The core of our interview questions focuses on individuals' cognitive processes in their analytical roles in incident detection and investigation, parallel to our research questions. It explores how data is utilized in their roles, the data-related challenges they encounter during incident detection and investigation, cognitive hurdles, factors contributing to cognitive overload, and their experiences with automation and AI, including its benefits and limitations during their daily routine of data triage, pattern recognition, critical thinking, hypothesis generation, and problem-solving. Through these discussions, we aim to gain insights into the

main cognitive elements and understand how they collectively shape the analysts' approach to their tasks. The full list of the interview questions is included in Appendix 1.

4.2 Participants

We recruited a total of 18 participants through cybersecurity conferences, LinkedIn, and local firms. All participants have working experience in cyber defense analysis and incident response. The participants come from companies across various industries, including Hospitality and Tourism, Technology/IT, Finance/Banking, Healthcare, and Manufacturing. We aimed to achieve a sample of professionals who have a similar set of job responsibilities that require active data analysis and incident detection, which are often different from high-level managerial positions. The distribution of roles and experience levels among our participants is detailed in Table 2. All but one participant (P9), who is a security analyst intern, are full-time employees in their respective companies. Although P9 is an intern, their responsibilities align with those of the full-time analysts in our study. With 9 months of experience in the role, P9 has sufficient exposure to the tasks and challenges relevant to

Table 2 Participants of Study 1

	Role	Experience	Security Team Size	Company Size	Industry
P1	Security Analyst	1–2 years	11–20 members	> 1000 employees	Hospitality and Tourism
P2	Security Researcher	3–5 years	> 20 members	> 1000 employees	Technology/IT
P3	Security Analyst	1–2 years	6–10 members	> 1000 employees	Transportation and Logistics
P4	Security Analyst	1–2 years	6–10 members	201–500 employees	Technology/IT
P5	Detection & Response Analyst	1–2 years	11–20 members	> 1000 employees	Finance/Banking
P6	Senior Security Engineer	> 5 years	1–5 members	> 1000 employees	Technology/IT
P7	Security Analyst	1–2 years	6–10 members	501–1000 employees	Technology/IT
P8	Threat Detection Analyst	3–5 years	6–10 members	201–500 employees	Finance/Banking
P9	Security Analyst Intern	< 1 year	6–10 members	51–200 employees	Technology/IT
P10	Security Incident Analyst	3–5 years	1–5 members	> 1000 employees	Healthcare
P11	Cyber Threat Analyst	1–2 years	21–50 members	> 1000 employees	Finance/Banking
P12	Incident Response Lead	3–5 years	> 50 members	> 1000 employees	Technology/IT
P13	Senior Security Engineer	> 5 years	21–50 members	> 1000 employees	Manufacturing
P14	Detection & Response Engineer	> 5 years	> 50 members	> 1000 employees	Technology/IT
P15	Security Engineer	1–2 years	1–5 members	11–50 employees	Technology/IT
P16	Security Analyst	4–5 years	6–10 members	> 1000 employees	Finance/Banking
P17	Detection Developer	2–3 years	21–50 members	> 1000 employees	Technology/IT
P18	Security Analyst	1–2 years	> 50 members	> 1000 employees	Technology/IT

our research objectives. This diversity in our participants' backgrounds and their respective industries ensures a balanced representation of views and experiences within the field of cybersecurity.

4.3 Data analysis

We transcribed the interview data using Zoom's automatic transcription function to obtain an initial draft of the interview content. To ensure accuracy and completeness, the transcriptions were subsequently reviewed and refined by the authors and a graduate student. This iterative process involved verifying the automated transcription, correcting errors, and ensuring the final text accurately reflected the participants' responses. In total, we processed and analyzed 438 pages of transcriptions. The interview transcriptions were analyzed using the thematic analysis method. We employed thematic analysis due to the design of our semi-structured interview questions, as well as for its inherent flexibility and adaptability (Belotto, 2018). The thematic analysis enables the systematic identification, analysis, and interpretation of patterns or themes within the diverse and rich responses elicited by our targeted questions. We followed a structured coding procedure (Charmaz & Belgrave, 2012): first, both authors participated in all the interviews and reviewed all interview transcripts. With a thorough familiarity with the data, the authors noted initial ideas. Subsequently, both conducted initial coding by highlighting text segments and assigning codes that represent their essence. The authors met to discuss the codes until an agreement was

reached. After initial coding, the authors grouped the codes and identified themes related to the ideas and concepts of interest and met on a weekly basis to review the themes, ensuring the themes were consistent throughout the dataset. Table 3 provides an overview of the code structure for this study. This method enabled us to efficiently uncover and link the underlying themes directly related to our research questions.

4.4 Key Findings

4.4.1 Data Challenges with the Advent of AI and Automation

A significant portion of our study focused on understanding the data challenges faced by analysts in their day-to-day operations. First, participants highlighted that the overwhelming amount of data, coupled with a high rate of false positives, leads to alert fatigue. For example, P2 pointed out that the transition of companies to the cloud and the digitalization trends contribute to the issue of data volume, "*A lot of companies move things to the cloud, and the cloud logs are much longer compared to normal, so just the size of the logs is very big, especially with cloud.*" This data challenge necessitates the implementation of data triage and alert correlation strategies. The adoption of AI and automation varies across companies, with some utilizing scripts for data triage and others employing machine learning to filter and correlate alerts. Six participants stated that they use self-defined scripts to pinpoint false alerts and manipulate data. Only

three participants acknowledged that they frequently work with machine learning detection results in their jobs. For example, P12 noted, *“We also use a lot from the vendors that we work with. Some use machine learning to filter alerts. I like higher fidelity machine learning detections versus more generic alerts.”*

Another significant challenge comes from the inconsistencies in data formats. The logs from diverse sources and vendors exhibit inconsistencies in format and variable naming, which poses significant challenges for analysts in data integration and interpretation. P7 highlighted this issue as follows, *“Services store and display that data and those logs differently and in their own unique ways. So you have to learn all these different ways to parse that data and understand that data and interpret that data.”* To standardize and consolidate data, some analysts employ self-constructed scripts to process data. Meanwhile, all participants mentioned that their companies leverage security information and event management (SIEM) systems to centralize data and facilitate data analysis. P3 noted, *“having everything consolidated into that SIEM tool allows us to be a lot more efficient with our investigations.”*

Furthermore, the lack of a comprehensive understanding of the enterprise system and the specifics of the collected data potentially leads to misinterpretations, oversight, or not knowing what to investigate. Analysts at different levels of experience mentioned the importance of getting familiar with the available data in the environment. P10 described it as follows, *“An analyst also has to know the environment.... because now you have way too many entries, way too much data flowing. ... just to actually make sure that you're looking at the right data.”*

Another interesting finding is the changing nature of the learning process. Traditionally, analysts learn on the job from senior colleagues. Senior analysts, with their extensive working experience, are more familiar with interpreting logs and recognizing patterns of attack behaviors. This knowledge transfer from senior analysts ensures juniors comprehend the subtle intricacies of their environment and tools rapidly. As generative AI tools become popular, however, junior analysts are now also leveraging these tools to interpret logs and rely less on senior analyst inputs. The majority of the participants mention they have been using generative AI tools to interpret logs or generate search queries. For instance, P12 stated, *“I also find the AI (Gen AI) is really helpful when I need to generate a query. For example, let's say I'm not familiar with Splunk logic. Then I can ask AI to write a query.”* Although the degree of AI adoption for this purpose varies across companies, it is a trend to ease the workload of senior analysts and create opportunities to scale and standardize the knowledge transfer process. However, participants also emphasized that AI cannot fully replace the business-specific or context-aware insights provided

by senior analysts, particularly when a deeper understanding of attacks or alignment with organizational priorities is required.

4.4.2 Investigation as a Dynamic Cognitive Process

When participants were asked to describe their detection and investigation process, they uniformly began by distinguishing between true alerts and false positives. Once they identified true alerts, the next step involved prioritizing these alerts and determining the immediate action to stop the spreading of the incident, a phase one participant described as “stop the bleeding,” aiming to stabilize the situation quickly. Following this initial response, the investigation typically deepens into a root cause analysis, which varies from case to case. A common approach is back-chaining, which refers to interpreting the current alerts and reflecting on the attacker's lifecycle to formulate hypotheses about the attacker's behaviors, guiding further investigation. The formulation of hypotheses based on the attack lifecycle is the key to enabling analysts to ask the right question, which also requires their domain knowledge and experience. For example, P1 described a method of working backward to trace the origins and intentions of the suspect activities: *“I'd work backwards and say, Okay, where were they coming from? ... getting more information to say these were malicious”*. P8 emphasized the importance of strategic thinking involved in these investigations: *“... you have to rack your brain as to if you were the attacker or the threat actor. How? What would you be trying to do? Where would you be trying to look, and what steps might you be taking to hide your activity?...”*

The following in-depth investigation is described as a cycle where analysts consistently formulate hypotheses about the attacker's behavior, seek evidence, and then make decisions on the next steps based on their observations. P2 iterates as *“I start to think about where that is in the attack lifecycle. In that example of Powershell downloading something ... so I ask ... How did that get there in the first place? And what might lead me to see it? Oh, a user got phished? And then they clicked the link and downloaded this file. Who else may have sent that link? ... so I ask every question, and the goal is to lead me somewhere in the attack lifecycle.”*

Similarly, P7 describes the process as: *“... So from that file hash, we would start searching for what is associated with it. If it's associated with something potentially malicious.. we'll start investigating the workstations and the logs on those workstations to see where this file hash came from, and we'll be sorting through the logs to just kind of try and get a step by step. How did this get on the system?... if it's from an email, let's say, who is the person who sent them? We will search emails and see if that's someone that they correspond with regularly. And if it's not, then again, that's something suspicious.”*

Table 3 Code structure in Study 1

Category	Theme	Initial Code
Data challenges with the advent of AI and automation	T1: Data challenges in incident detection and investigation	- Overwhelming data volume and false positives - Variation in data format
	T2: Growing importance of AI and automation in incident detection and investigation	- SIEM-enabled data centralization - Rule-based data triage - Machine-learning-based data triage - Employing scripts for data manipulation and response action
Investigation as a dynamic cognitive process	T3: Cognitive functions in alert triage	- Prioritize alerts based on impacts - Inspect alert details and payload - Confirm alerts by engaging in communication - Identify false positive
	T4: Cognitive functions in in-depth investigation cycle	- Reflect on the attack cycle - Construct hypothesis about attacker's behavior - Search for and collect evidence within logs - Interpret observations - Collect evidence - Maintain the investigation flow
Cognitive challenge in dynamic investigation processes	T5: Cognitive challenges in alert triage	- Information overload - Repetitive and tedious work - Reduced alert sensitivity from frequent false positives
	T6: Cognitive challenges in the in-depth investigation cycle	- Lack of visibility - "Rabbit hole" problem - Creative thinking in high-pressure situations
Insights on AI and automation	T7: Limitations of current AI and automation	- Concerns about the accuracy of AI-driven data triage - Lack of trust in AI results
	T8: Expectations for future developments	- Expectations for more AI applications in data triage - Expectations for AI explanations

To sum up, when describing the analytical processes in incident detection and investigation, participants noted that their tasks vary from case to case due to the complex and uncertain environments in which they operate. The analytical process is identified as a dynamic cognitive process that involves evolving perceptions, attention, memory, reasoning, and decision-making. Specifically, the investigation process consists of iterative cycles, each of which involves identifying true alerts, taking immediate containment actions, conducting further root cause analysis by generating hypotheses about attacker behavior, and seeking evidence to validate the hypotheses. We refer to an iterative cycle as an *investigation flow*, which corresponds to the analytical cycle described in the conceptual AOH model. In this model, analysts conduct

hypothesis-driven actions, obtain observations, and draw new or updated hypotheses (Zhong et al., 2017).

Consequently, we categorize the data analytical operations into two main stages to better understand the impact on cognitive overload (RQ2): alert triage and in-depth investigation cycle. Alert triage is the initial stage where analysts differentiate between false positives and true positives, aiming to filter out the false positives and escalate the true positives. The second stage, in-depth investigation, involves an interactive cycle of data search, evidence collection, hypothesis formulation, and deciding on the next steps of the investigation. The next section presents the significant cognitive challenges faced by analysts in each of these major stages, providing insights into the complexities of their roles.

4.4.3 Cognitive Challenge in Dynamic Investigation Processes

Our analysis showed that information overload is one significant cognitive challenge in the alert triage stage. With the prevalence of false positives and the overwhelming volume of data, analysts, particularly those who are less experienced, face a daunting task. As P2 pointed out, *“security analyst is very high burnout, not because the work requires a lot of thinking, but because it is voluminous and repetitive.”* P1 noted, *“I’m so burned out from working on 10 alerts that I know are false positives.”* Parallel to the prior discussion on Information Processing Theory, the working memory has a limited capacity, and junior analysts are likely to struggle more due to their unfamiliarity with the alerts and data, affecting their efficiency in processing and interpreting information (Roetzel, 2019). In contrast, senior analysts, armed with both experience and a developed “taste” for distinguishing between alerts, are adept at identifying which alerts need further investigation and which can be categorized as false alerts more efficiently. In other words, because senior analysts rely on existing mental models in their long-term memory more frequently, they are able to dedicate more time to the deeper, critical reasoning required in subsequent stages of an investigation. However, even senior analysts can find their work monotonous and lacking in variety. P2 noted, *“it was exciting to see the rare alerts.”* In addition, the time pressure further contributes to cognitive overload. As P12 stated, *“sometimes the alerts have time restrictions with contracts... we just don’t have enough time when there’s 200 other customers that also need alerts triage.”*

When participants described their methods for distinguishing between true alerts and false positives, they used a combination of problem-solving strategies, including memory retrieval, analytical (exploitative), and creative (exploratory) strategies (Albanese et al., 2017). They frequently referenced the use of heuristics, aligning with the discussion of Dual-Process Theory (Evans, 2008), which defines System 1 and System 2 thinking. System 1 involves intuitive, fast, and automatic reasoning, while System 2 encompasses more deliberate and rational thought processes. In the high-pressure environment of incident detection and investigation, participants confirmed that they often rely on System 1 thinking, leveraging their intuition to make quick decisions. For example, when commenting on alert triage, P8 noted, *“When you’re faced with an easy problem, after being in the field for a year, you almost reach an autopilot state, where, if you see the same kind of activity 10–15 times, you’ve already gotten accustomed to how to handle that, what steps need to be taken, it becomes a very clear and precise standard operating procedure.”* This reliance on intuition underscores the importance of experience in the field, as senior analysts develop a keen sense of identifying

potential threats. For instance, P2 commented, *“I’ve developed a pretty good instinct for something that’s probably bad.”* These comments illustrate that intuitive judgment plays an important role in their daily responsibilities. However, participants also acknowledged that this reliance on intuition can sometimes lead to oversights. A common way to overcome this is to get team members to review the alert. All participants mentioned that collaborating with others in the team is very frequent.

When intuitive reasoning falls short, analysts turn to System 2 thinking, engaging in a more thorough and systematic evaluation. However, System 2 thinking brings its own cognitive challenges, particularly the fear of missing critical information, articulated as *“Did I really look inside enough?”* This concern can lead analysts down a “rabbit hole,” where they invest excessive time and effort into an investigation, gathering more information than necessary. For example, P2 commented, *“junior analysts tend to spend too much time...some questions can be addressed later... Most customers don’t care about a lot of the details. They just want to know what happened and what they need to do”*.

Such scenarios highlight the issue of attention bias, where the focus on certain details may detract from the broader context. These scenarios may also result in a delay in decision-making when an analyst becomes so overwhelmed by the process of analyzing and overthinking decisions that they become unable to take action. P4 shared an example, *“The activity was rare to me, I found myself going down rabbit holes, reaching all the details, coming up with various possibilities. It took time, but I couldn’t verify any of them”*. Our analysis showed consensus on how focusing too much on in-depth analysis in high-pressure situations, especially when quick, decisive action is needed, can hinder effective response to a situation.

Based on participants’ feedback on the challenges of triaging false positives, we also noted the “Cry Wolf” effect: As analysts face repeated false alerts, they may become less sensitive to incoming alerts. Consequently, the decreased sensitivity can result in a higher likelihood of missing genuine signals. This challenge is associated with the extraneous load in CLT. For example, P2 mentioned, *“if they (analysts) get nothing but false positives, and then the day that something real real bad happens, maybe they’re just not going to trust that.”* It is a critical challenge in maintaining vigilance and accuracy when the false positive ratio is high and the cost of missing a true signal is high. Some participants mentioned that missing true positives may cause significant stress and negatively impact their emotions.

During the in-depth investigation cycle, analysts delve deeper into the logs to identify the root cause and understand the attack behaviors. According to the Dual-Process Theory, this stage often involves both System 1 and System 2 thinking. For straightforward problems where the incident

type is familiar and sufficient information is accessible, analysts employ heuristics, utilizing System 1's quick and intuitive reasoning. For example, P5 commented, "... (for the easy problems) the main thought process is just following those playbooks. So the same methodology to analyze, contain, and understand how it happened... You just sort of repeat the cycle until you have a complete understanding." P10 remarked, "if it's malware, how did they get on the machine... you start asking these questions. And then over time, you just do it. Cause you've done so much."

In these situations, analysts quickly navigate through the steps without significant cognitive effort. However, participants overwhelmingly agreed that the difficulty of multitasking and the nature of their work demands deep concentration to maintain the flow of investigation. A consistent theme that highlights the prevalent interruptions in the investigation includes communications with customers (messages and calls), on-the-fly conversations, scheduled meetings, and other side tasks. P10 noted, "*one challenge is lack of investigation time, especially when you have multiple hats. Sometimes I have back-to-back meetings, I may need to jump on a meeting before finishing the investigation.*"

In contrast to the scenarios driven by System 1 thinking, analysts frequently encounter significant cognitive challenges when reasoning under uncertainty, particularly when there is a lack of contextual information. This situation often arises due to a lack of visibility of the customer's system or when analysts face activities that are rarely known or documented. In these instances, analysts need to construct a mental model of the incident with the limited data available. They need to adjust to the gaps in data and depend more on abstract thinking and hypothesis generation. This more complex cognitive process demands a higher level of analytical reasoning, often involving System 2 thinking, to piece together the puzzle of the incident with whatever information is available. Therefore, creativity is the key, which is also described by participants as the ability to "*think outside of the box.*" For instance, P6 commented on creative analytical strategies saying, "*harder problems require more cognitive skills. So, thinking outside of the box, like different searches, different ways to think about the problem, if it's not caused by XYZ, could it be caused by ABC, so there is a different analytical strategy of how you approach the problem.*"

A common challenge highlighted by participants is the lack of clues, where analysts may find themselves without a clear next step. This can sometimes be due to a shortfall in creative thinking or a lack of contextual information. Most of the participants stated that teamwork plays an essential role in navigating this challenge. P11 described their team collaboration, "*When there are no clear next steps, everybody on the team is super open to help out. I can easily ask a domain expert to review my work and suggest what to do next or what data to check. I believe that is definitely one of*

the strengths of our team." In addition, participants stated that they have been using Generative AI tools to get investigation ideas. For example, P11 described how an internal AI tool can be used for this purpose, "*I can ask it [the Generative AI tool] what other behavior should I be looking for? And it can tell me some attack behaviors that have been seen and suggest what I should look for next. For example, [it suggests that] you could look at the surrounding processes, or you could look at the users logged in. And I could further ask what data can tell me user login information... it is really incredible.*"

Moreover, the impact of stress on cognitive processes is notably significant, particularly during the intense in-depth investigation cycle. On one hand, complex problems demand thorough sense-making and verification of information. On the other hand, analysts are under immense pressure because of the limited investigation time, potential serious impacts of the incident, and urgent needs of waiting customers. Therefore, all participants indicated that engaging in deep analytical System 2 thinking becomes challenging due to the significant cognitive overload. P12 described the difficulty of maintaining concentration during serious active incidents, especially when the situation is still unfolding and immediate action is required, "*...when you're dealing with a really serious incident, especially when an incident hasn't been stopped yet and we haven't fully mitigated everything... every second counts, so we have to concentrate... In those stressful situations, it is tough to concentrate. I found it's especially hard when customers are asking questions...*"

Documentation adds another layer to the cognitive load. Given team collaboration is common, "*warm handoff*"—a process where analysts' findings and contextual information (e.g., analytical strategies, tools, evidence, and unproven hypotheses) are transferred seamlessly between team members—is essential for cognitive continuity and efficiency. P5 highlighted the importance of documentation, "*So I need to give them all the details on highlights of what's been done and what needs to be done to someone else, so they can look through my work and continue the investigation.*"

4.4.4 Insights on AI and Automation

Our analysis revealed that all participants have some level of experience with AI and automation. When asked about the use of AI, their discussions predominantly centered around generative AI tools and illustrated the growing role of these tools in their jobs. The primary applications of AI highlighted by participants include constructing scripts for data manipulation, formulating commands, constructing search queries, facilitating knowledge inquiry, and log interpretation (Refer to the relevant codes in Table 3). However, when probed specifically about their working experience with learning-based AI detection tools, many participants

indicated either a lack of experience with or unawareness that the tools they were using incorporated learning-based AI models. This finding is intriguing, considering the extensive academic exploration of learning-based AI intrusion detection systems over the past decades (Kocher & Kumar, 2021). Rule-based detection systems still dominate in the practical domains. Analysts prefer the fact that the rule-based detection alerts are interpretable; they can review the rules to understand why alerts were triggered. Participants expressed significant concerns regarding trust in learning-based AI outputs, especially when these outputs do not provide information about the factors triggering the alerts. For example, P11 noted, “... when I look at them [rule-based alerts], I trust them because they're high fidelity and the rules tell me how they work. If you don't understand how the alert is generated, you can't trust it”.

Participants acknowledged the growing role and promising potential of AI in enhancing their job efficiency. However, they also highlighted several limitations of AI, including the challenges in verifying its decisions when errors could have substantial consequences, its inability to account for contextual information like human relationships and social events, and limitations in understanding attacker behavior, particularly in the case of novel or creative attacks. Consequently, all participants viewed AI as playing a supportive role that augments human cognition. They expressed a strong interest in leveraging AI for data triage to address the data challenges. To enable more effective collaboration with AI, they advocated for making AI systems more transparent and ensuring acknowledgment of AI-generated results.

During the interviews, we described a scenario using machine-learning-based detection and asked the participants whether and when they would want to read the explanation of the machine-learning model. We observed the evolving nature of analysts' informational needs from participants' responses. The majority of the participants expressed their willingness to review such explanations, indicating they are always open to additional information. However, when asked about when they would integrate this information into their investigation flow, only one participant (P7) mentioned they would do so during the investigation, while all others preferred to review it at the beginning. P7 describes these moments, “I may need to see [the information about how the machine learning model was trained] during the investigation when I get lost. I may also want to read them all at the end of the investigation to further verify my findings. I would be able to determine if the AI was accurate. I may also want to check it out probably the first 5 min of investigating the alert to scan over it and see if it [detection alert] is accurate?” P7 also believes that explanations

can be helpful but pointed out that they could be disruptive if they provide incorrect information, stating, “The only way that I could really imagine it being disruptive is if it was completely wrong and the analyst doesn't realize or verify it. Otherwise I think explanations would only be helpful.”

The dichotomy between the desire for more information and the practical limitations further underscores a critical aspect of analysts' work: the need for uninterrupted focus. Given the cognitive challenges mentioned above, maintaining a continuous chain of thought is essential for effective analysis and decision-making. Participants identified that the introduction of information that is not directly related to the current task could also pose a considerable disruption. P12 emphasized the importance of pertinent information, “As an investigator, I don't really care how the (machine learning) model was trained. Moreover, I care if it's accurate, or if it tells me what I need to know, to understand the attack behavior.”

5 Study 2: A Case Study

5.1 Study Background and Goal

Participants in Study 1 expressed a generally positive outlook toward AI integration in their workflows but voiced significant concerns about AI system reliability. Discussions about AI during interviews consistently converged on one critical point: the necessity of explaining AI results to enhance understanding and trust and enable verification of AI-generated results. Additionally, the interview study highlights the evolving data and cognitive challenges faced by cybersecurity professionals with the advent of AI and automation. More importantly, our findings illustrate that these challenges will directly impact analysts' information needs during an investigation process, which in turn will directly influence their use of AI explanations in the process.

Motivated by these findings, the second study is designed to investigate further into individual investigation processes, aiming to understand how analysts' need for AI explanations emerges, evolves, or disappears throughout these processes. In Study 1, participants indicated having experience in varying levels and with different types of AI tools. In Study 2, by conducting an experiment in a controlled environment, we ensure all participants use the same AI tool and work on the same data set of alerts. In other words, while Study 1 revealed the perceptions and expectations of the participants, Study 2 gave us the opportunity to capture the participants' actual experience with XAI in an investigation task. This exploration in Study 2 provides a more granular understanding of our final research question.

5.2 Study Design

In this case study, we examine the behaviors of cybersecurity professionals as they work on a simulated incident response task, designed to mirror the complexities and challenges of real-world cybersecurity investigations. We aim to gain insights into the data triage and in-depth analysis processes and the need for AI explanations. To this end, we recruited six new participants with working experience in cybersecurity. To avoid potential bias, Study 1 participants did not participate in Study 2. Prior to the study, the participants were screened based on their understanding of common cyberattack events and cybersecurity investigation procedures. After the screening, the participants were asked to complete the investigation task within one hour. Through a detailed analysis of their behavior during the task, this study aims to uncover the dynamic need for AI explanations during the investigation processes.

We denote the participants in this study as A, B, C, D, E, and F. Participants A, B, C, and D have less than one year of experience in incident response, E has 1–2 years of experience, and F has 2–3 years of experience. Additionally, the participants came from a variety of industries, including IT, consulting, and healthcare, offering a broad spectrum of perspectives. Specifically, participant A worked in healthcare, participant B in consulting, and participants C, D, and E in information technology. While the sample size may limit the generalizability of the findings, the insights gathered are informed by diverse real-world contexts.

5.3 Task Description

The incident response task provided participants with network traffic data and a number of AI-generated alerts. The network traffic included three underlying attack events. Participants were tasked with uncovering and reporting these attack events by the task's conclusion. The network traffic data was extracted from the MSCAD dataset (Multi-Step Cyber-Attack Dataset) (Almseidin et al., 2022), which was compiled from a multi-step attack on a simulated enterprise network.

The AI-generated alerts were produced by our self-built Random Forest classification model using Scikit-learn (Pedregosa et al., 2011), which was trained using the MSCAD dataset. Each alert is a prediction made by the model. We further utilized the LIME (Local Interpretable Model-agnostic Explanations) Python library (Ribeiro et al., 2016) to generate explanations for each alert. These explanations included the probability of prediction outcomes, the impact of features contributing to the prediction, and the values of these features for the current alert, all visualized through LIME. An example of such an explanation is illustrated in Fig. 2.

To facilitate interaction with the alert explanations, we developed software using Python. This software simulated a cyber incident response dashboard and displayed alerts to participants. It also included an “Explain” button next to each alert to request an explanation for the given alert. Figure 2 displays the user interface of this software. Before the task, to enable participants to engage with AI explanations effectively, we gave each participant a tutorial that explained the details of LIME and ensured participants were proficient in interpreting LIME outputs and understood how to request this information during the task.

At the beginning of the task, we informed participants that the alerts were produced by the machine learning algorithm and included false positives and duplicate alerts. During the task, participants were encouraged to request explanations for any alert as needed, with no restrictions on the number or timing of requests. After obtaining informed consent, we observed the participants' behavior (“shoulder surfing”) and recorded their screens to capture all their analytical and note-taking activities. This approach was designed to minimize disruptions. At the end of the task, participants were required to submit a detailed incident report. This report was expected to outline their findings and include specific evidence gathered from the data analyzed during the task. After participants completed their tasks, we interviewed them to clarify their behaviors during the task.

5.4 Key Findings

All participants managed to uncover the three attack events. Figure 3 provides an overview of the operation sequences performed by participants during the task. The operations include SEARCH (searching data), EXPLAIN (requesting AI explanations), BROWSE (reading an alert), and HYPO (noting down hypotheses about the potential attack event). We focus on how they use AI-generated alerts and the impact of alert explanations in data triage and in-depth investigation stages.

5.4.1 Individual Differences in the use of Cognitive Functions

A key observation from our study was the prioritization strategy employed by participants when triaging data. As displayed in Fig. 3, all participants, except for Participant D, began their investigation by identifying alerts of interest. In contrast, Participant D first checked the statistical description of the network data (e.g., the frequency of IP addresses, ports, and the different types of network connections) and then prioritized network events based on his understanding of abnormal events. Therefore, he started by searching based on network event type. This observation is consistent with the Study 1 interview finding that the participants allocated

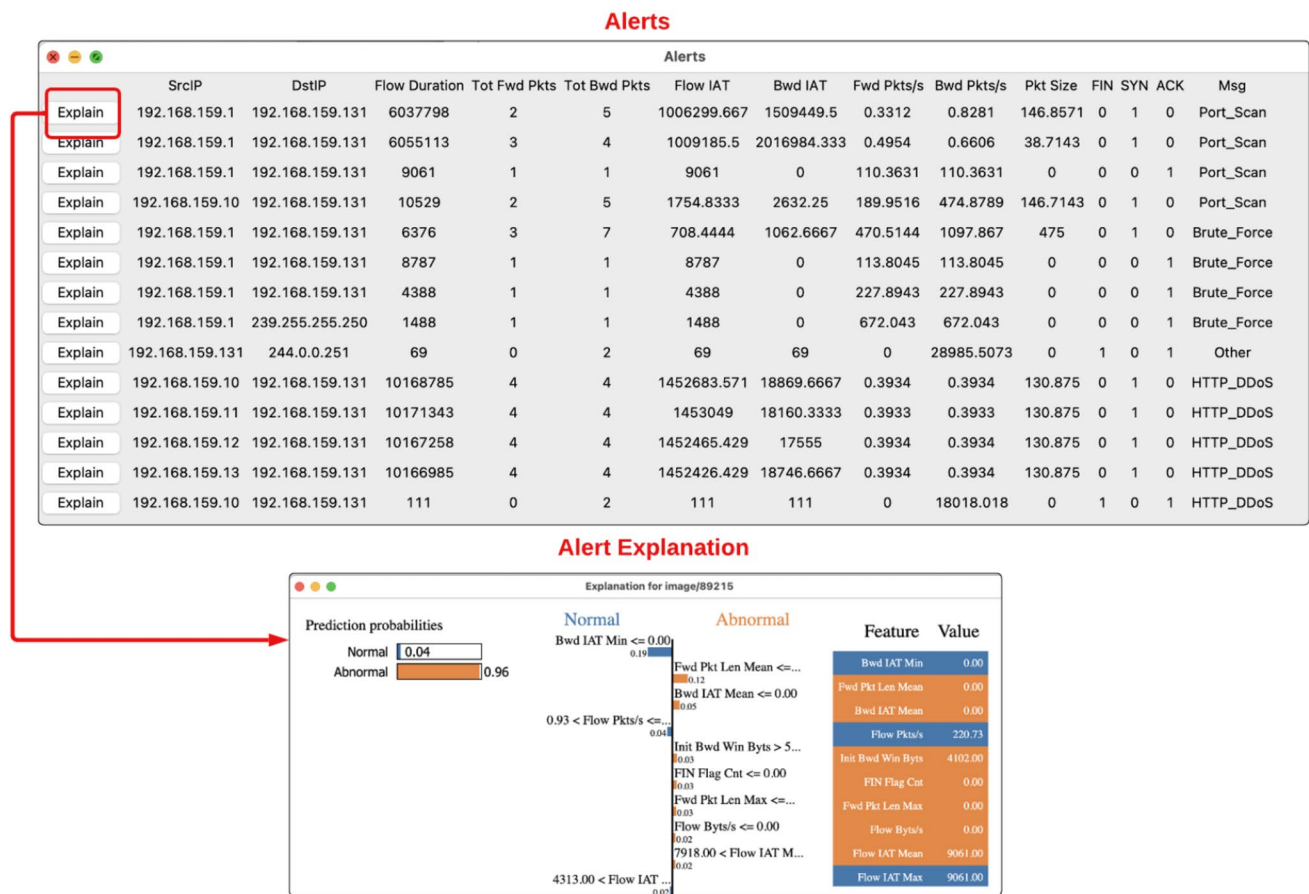


Fig. 2 The user interface of the software that was developed to display alerts and alert explanations

their attention and time based on perceived importance or relevance.

We observed that analysts maintain investigation flows during their in-depth analytical cycles. According to Theme T4 in Study 1, we observed that participants maintain their investigation flow at various moments, as their task operations are goal-oriented, typically involving hypothesis generation and evidence gathering to validate these hypotheses in a continuous manner. The “gaps,” or idle moments, where no actions are taken, often occur after completing the investigation of one hypothesis and before addressing a new target for investigation.

5.4.2 Variability in Participants' use of AI and AI Explanations

We further investigate the variations in how participants utilized AI-generated alerts and AI explanations throughout their investigative processes. While most participants leveraged AI-generated alerts to pinpoint targets for subsequent investigation, notable individual strategies emerged. For instance, apart from Participant D, when participants

engage with AI-generated alerts, they commonly review explanations about the AI's confidence level and discern critical features to determine if they trust the alert. Once a true alert is identified, the alert details further inform their subsequent search for supporting evidence within the network data. However, the usage of AI and its explanations wasn't monolithic. Participant A, for example, also reviewed alerts after completing the investigation of a hypothesis to ensure thoroughness. In contrast, Participant D predominantly referenced alerts and their explanations towards the investigation's end, serving as a final check against the gathered evidence. After completing the tasks, all participants acknowledged the value of explanations in improving their understanding of alerts and aiding their decision-making process.

Participants didn't conform to a singular approach during the in-depth investigation stage. Rather, they adopted a variety of methods tailored to their investigative needs. For instance, Participant B initiated investigations by filtering the network data by IP addresses mentioned in the alerts, followed by a comprehensive review of all network traffic associated with those IP addresses. This method allowed

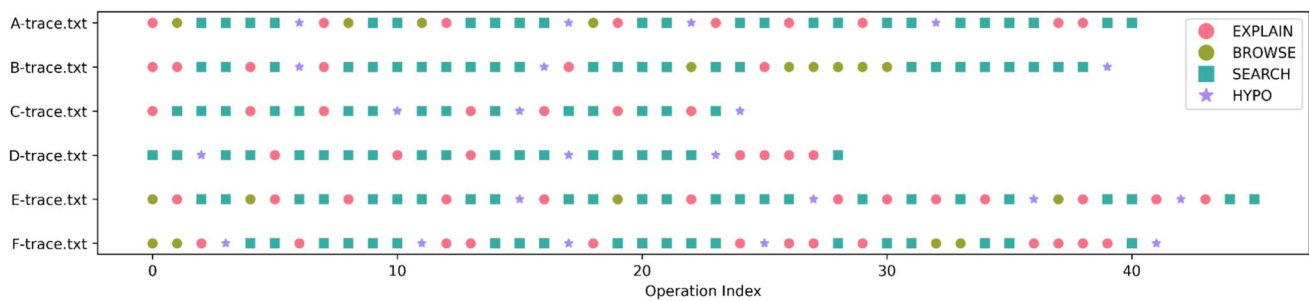


Fig. 3 Visualization of participant operations ordered by time. “Operation Index” denotes the i -th operation conducted by the participants

Participant B to narrow his focus to the relevant network traffic systematically. Hence, Participant B consistently reviewed alerts and their explanations before embarking on a new investigative cycle. Conversely, Participant F preferred to correlate multiple alerts to identify recurring IP addresses. Therefore, he often reviewed several alerts and their explanations simultaneously. These observations reflect the diverse cognitive approaches used by cybersecurity professionals and underscore the varied ways in which AI tools and explanations are integrated into investigative workflows.

5.4.3 AI Explanation may cause Cognitive Challenges

Our observations and post-task interviews highlighted a paradoxical effect: while AI explanations are designed to facilitate the analytical process, they may inadvertently introduce cognitive challenges that complicate the analysts' work. According to Study 1, the “rabbit hole” problem emerged as a significant cognitive challenge among analysts. In this study, although not mandated, every participant engaged with the AI explanations, driven by a fear of missing out on crucial information. This engagement, however, sometimes led them down time-consuming paths of investigation that were not always fruitful. For example, Participant E encountered an AI explanation highlighting a TCP“FIN”flag as a significant feature for the alert generation. This information propelled the participant to delve into an extensive search for TCP traffic with the“FIN”flag. This pursuit expanded the scope of the investigation but didn't provide definitive leads. Despite the additional effort, this line of investigation did not yield valuable evidence. The participant had to return to the initial alert and start a new investigative cycle focused on IP addresses. The additional time and effort spent navigating these“rabbit holes”can increase the cognitive load on analysts, potentially leading to burnout. Participant C expressed this concern at the end of the task, “*it is likely to burn out an analyst in a real-world case when there are thousands of alerts.*”

The above observation has indicated that integrating AI explanations into the investigation workflow is not

straightforward. Given that all participants tend to check explanations due to the fear of missing out, another cognitive challenge posed by AI explanations is the potential for overreliance. This can lead to a decrease in critical thinking and underutilization of the analysts' own expertise and experience. Reflecting on the above situation, Participant E points out that he relied too heavily on the explanations but did not fully leverage his analytical skills.

6 Discussion

6.1 Propositions for Designing XAI Systems to Address Cognitive Challenges

We focus on the individual cognitive dynamics during the cybersecurity investigation process. Study 1 examines the cognitive functions and challenges of analysts during the data triage and in-depth investigation stages and highlights issues such as cognitive load and bias. Study 2 captures analysts' real experience with XAI during an investigation task and expands on Study 1's findings by examining how analysts' informational needs, especially regarding AI explanations, evolve in response to the cognitive challenges encountered during the investigation process. Together, these studies offer a comprehensive view of the interplay between cognitive challenges and the evolving requirements for AI explanations. We identify four principles for addressing the cognitive challenges of XAI based on the findings of these two studies, as summarized in Table 4. The first two principles emphasize the need for XAI systems to adapt to analysts' goals and cognitive load, ensuring that AI explanations are relevant, effective, and do not contribute to cognitive overload. The third principle focuses on striking a balance between the depth and simplicity of AI explanations, enabling users to comprehend critical insights without being overwhelmed by unnecessary details. The fourth principle highlights the importance of considering users' expertise and preferences and ensuring dynamic interaction. Based on these principles, we further present six specific and testable

propositions for addressing cognitive challenges of XAI in the incident detection and investigation workflow, as detailed below.

Study 1 participants highlighted numerous data challenges and emphasized the need to sift through vast amounts of data while under constant pressure not to overlook any critical details, particularly during time-sensitive and critical incidents. To address these data challenges, they expressed expectations for employing AI in data triage and enhancing the transparency of AI. Meanwhile, participants in Study 2 found that AI explanations aided their understanding of AI-generated alerts. These findings highlight the potential advantages of AI explanations in enhancing analysts' understanding of AI-generated results. However, the range of cognitive challenges uncovered in Study 1, combined with concerns that AI explanations could potentially overwhelm analysts with excessive information, necessitates a careful balance. Given the dynamic nature of the investigation process, with its evolving cognitive loads, analysts' needs for explanations change over time. The timing of the presentation of AI explanations could effectively impact cognitive load. In both studies, we observed that the in-depth investigation cycles are highly goal driven. Reflecting on the insights from Study 1, an investigation cycle usually starts once analysts identify a target and form a strategy for investigation. Moreover, analysts need to concentrate on the goal during the investigation cycle. Therefore, introducing AI explanations as additional information during such a cycle could potentially disrupt the investigation. Accordingly, our findings in Study 2 revealed varied preferences in the timing of requesting alert explanations. Participants requested explanations at the onset of investigating each alert, using this information to determine the likelihood of false alerts and assist in their initial decision-making. Participants also opted to seek explanations towards the end of their alert investigation to confirm their findings. Few participants requested explanations midway through their investigations since explanations could be disruptive or even lead to a reassessment of decisions. Accordingly, we present the following proposition:

Proposition 1: XAI systems that adapt the timing of explanations based on a real-time understanding of analysts' goals (e.g., prioritizing alerts, gathering evidence, or verifying results) will enhance the relevance and effectiveness of the explanations.

Additionally, according to the first two principles in Table 4, XAI systems must account for the analysts' goals and cognitive load across various stages of the incident detection and investigation workflow. This requires dynamically adapting explanations based on the current context and ensuring alignment with the analysts' specific goal at each

stage. This complexity leads us to propose the following two propositions:

Proposition 2: XAI systems that estimate cognitive load in real-time using indicators such as task complexity, user interaction patterns, or physiological signals (e.g., eye tracking or response time) will be better equipped to adjust explanation depth and format dynamically, preventing analysts from feeling overwhelmed.

Proposition 3: XAI systems capable of adapting to analysts' goals and real-time monitoring of cognitive load will improve task performance.

Participants in Study 1 expressed the fear of missing out and potential confirmation bias. Study 2 further provides insight into the cognitive challenges posed by AI explanations in investigation. The findings indicate that analysts are motivated to engage with the details provided in AI explanations. To overcome the potential information overload brought by AI explanations, AI explanations need to be carefully presented to balance the detail and depth of explanation details and ensure the presented contents are relevant to the current task. Therefore, we present the following proposition:

Proposition 4: XAI systems that prioritize presenting high-fidelity and critical information while allowing analysts to request additional details will help prevent analysts from becoming distracted or overwhelmed by unnecessary exploration, effectively avoiding the pitfalls of “rabbit holes.”

Study 1 findings suggested professionals' expectations for acknowledging the role of AI and the potential negative impact of AI explanations and emphasized the issue of over-reliance on AI results when AI explanations are available. This concern was further evidenced in Study 2, where we observed a participant excessively trusting the features highlighted in an AI-generated alert, leading to an inefficient allocation of investigation time. Recalling participants' emphasis on critical thinking and the application of domain expertise from Study 1, it becomes crucial to encourage analysts to leverage their own experience and domain knowledge when working with AI-generated results. One approach is to facilitate analysts in providing feedback on AI outputs easily. Given the existing burden of incident documentation, it's vital to create a streamlined process for analysts to record their insights, questions, and rationale without adding to their workload. Additionally, offering analysts the choice to opt in or opt out of engaging with AI can empower them to make informed decisions about when to rely on AI and when to lean on their expertise. Thus, Propositions 5 and 6 are presented as follows:

Table 4 Principles for addressing cognitive challenges of XAI in CSIR

Principles (What?)	Rationale (Why?)	Solutions (How?)	Related Propositions
Prioritizing relevant information to align explanation with analysts' investigation workflow	- According to the Information Processing Theory, analysts' working memory has limited capacity. Less important details may distract analysts and lead to unnecessary exploration - Analysts at different stages of an investigation may prioritize explanations differently based on their relevance and importance to the specific phase of the process	- XAI systems should highlight the most critical features that influenced the AI's decision - XAI systems should be seamlessly integrated into the existing investigation workflow, considering the different decision-making stages involved	P1, P2, P3
Offering contextualized explanations	- Analysts constantly form hypotheses about the possibilities of attack activities based on the information provided - Cognitive Load Theory suggests non-relevant information increases extraneous load. Analysts may go down tangential paths if the information in explanation is not relevant to the current investigation context	- Explanations should be updated dynamically as the investigation progresses - Explanations should be tailored to the context of the investigation and aligned with the analyst's goal in the given scenario	P1, P2, P3
Balancing depth and simplicity to manage cognitive load	- Dual-Process Theory suggests that analysis paralysis occurs when System 2 is overwhelmed with evaluating an excess of information. Too many details in explanations may increase the risk of analysis paralysis - However, analysts may prefer having the option to delve deeper into details when necessary	- XAI systems must present information in a concise and digestible format - XAI systems should provide explanations at different levels of detail, allowing users to access only as much information as they need without diving too deep unnecessarily	P4, P5
Encouraging user-centered design and dynamic interaction	- By aligning system design with user expertise and preferences, systems can better support System 1 intuition and System 2 analytical reasoning - Dynamic interaction provides flexibility to dig deeper or shift focus	- XAI systems should adapt to specific expertise and preferences of users - XAI systems should allow analysts to drill down into specific areas of investigation without losing sight of the overall investigation - XAI systems should provide analysts with the option to opt in or opt out of engaging with AI	P1, P2, P3, P4, P5, P6

Proposition 5: XAI systems that provide analysts with the flexibility to opt-in or opt-out of engaging with AI explanations will empower analysts to maintain control over the decision-making process and reduce overreliance on these systems.

Proposition 6: XAI systems that integrate human-recorded insights and questions into the learning process of AI models will lead to more adaptive and context-aware explanations over time.

6.2 Theoretical and Practical Implications

The propositions presented in this study highlight critical insights for designing AI and XAI tools for CSIR, emphasizing the need to accommodate analysts' varying situational awareness requirements. By synthesizing these design propositions with findings from our studies, we contribute significantly to the literature on cyber SA and XAI. Although a growing body of research has explored the impact of XAI in cybersecurity, a more nuanced understanding of how XAI affects analysts' cognitive processes is lacking. This gap was highlighted by previous studies revealing that XAI tools were often underutilized by analysts or deemed ineffective in practice (Nyre-Yu et al., 2022). To the best of our knowledge, this study is the first to specifically examine analysts' cognitive processes within the context of CSIR while interacting with AI systems and AI-generated explanations. Focusing on analysts specializing in incident investigation and response, our research provides detailed insights into how these professionals navigate and interpret complex and high-pressure situations. The findings extend existing cyber SA literature through the lens of CLT, shedding light on the cognitive strategies that analysts employ during decision-making in dynamic environments. Furthermore, our work emphasizes the crucial role of situational context in the effective implementation of AI and automation in cybersecurity. Notably, our research demonstrates that AI support and the nature of AI explanations must be tailored to the analysts' situated cognition, which varies with the context of their tasks.

Our study contributes to the growing XAI literature as well. While the extant body of XAI literature emphasizes the importance of explainability in terms of trust (Hoffman et al., 2023; Hsiao et al., 2021), transparency (Adadi & Berrada, 2018; Biswas et al., 2024), human-AI performance (Bauer et al., 2023; Holder & Wang, 2021), and regulations (Merhi, 2023; Yayla et al., 2022), design and usability of these XAI tools from the user's perspective have been neglected (Nadeem et al., 2023). Calls for more human-centric design of XAI tools underline the importance of considering users' goals, experience, and workflow for the successful adoption of these tools. Especially compared to

more traditional rule- and heuristics-based incident detection tools, AI-based tools do not provide an explicit cause of an alert. While explanations may help determine how the AI has made its decision, these explanations may in fact hinder the analyst's decision-making process by limiting their working memory and increasing cognitive load. Our study contributes to the limited findings in the literature by demonstrating how, in practice, explanations impact the decision process. Through the application of XAI in incident detection tasks in Study 2, we showed that analysts used explanations for different purposes. While some used it early in their analyses for *exploration* purposes while strategizing their following steps, others used explanations as *confirmation* for their findings. One of the interesting findings is that while participants felt comfortable using Generative AI in their decision process, they were wary of using learning-based AI, raising concerns about trust and fidelity, highlighting how the type of AI and the context it is being used affect an individual's perception of their benefits. Through our design propositions, we respond to studies that call for a more human-centric XAI that is designed and implemented with the consideration of its stakeholders in order to increase their adoption (Alahmadi et al., 2022; Liao & Varshney, 2021; Nyre-Yu et al., 2022).

We also contribute to practice by addressing the current analyst burnout and turnover challenge in the industry. Studies and industry reports consistently highlight that one of the primary challenges in CSIR is not merely technical but managerial. Specifically, finding and retaining skilled analysts has become increasingly difficult for organizations (Veilberth et al., 2020). The fast-paced, high-stress environment of CSIR often leads to burnout among analysts, contributing to high turnover rates. Industry reports indicate that the overwhelming volume of data and the sheer number of security alerts are key contributors to the increased workload and stress experienced by analysts. While AI-powered CSIR tools hold the potential to alleviate some of these challenges, the impact of XAI on analyst performance remains uncertain. Our findings revealed the importance of tailoring XAI tools to analysts' specific information needs to avoid unintended consequences, such as increasing their workload or prolonging decision-making processes. The design principles and propositions presented in this study aim to address these issues by reducing burnout and, consequently, turnover rates. Lower turnover rates would yield significant benefits for overall CSIR effectiveness, as retaining experienced analysts preserves institutional knowledge and expertise. In contrast, frequent hiring and onboarding of new employees result in disruptions and additional resource expenditures. Our research contributes by offering actionable insights for the development of AI-powered tools for effective CSIR practices and the well-being of cybersecurity analysts.

Another key contribution of our study to practice is in guiding the design of more effective XAI tools. Research consistently demonstrates that AI-powered tools outperform traditional rule-based or heuristic approaches in detecting cyberattacks. However, despite these advantages, the adoption of AI and XAI tools in CSIR has been slow. Organizations remain cautious about integrating these technologies, often perceiving them as delivering false promises (Nyre-Yu et al., 2022). Furthermore, industry reports reveal that over half of analysts believe their current CSIR tools increase their workload, which has been a significant deterrent to further investment in AI-based solutions (Vectra, 2024). Our findings suggest that for XAI tools to be effective, they must be tailored to meet the specific needs of analysts. Failure to customize these tools risks hindering analysts' performance. Without addressing these challenges, the implementation of XAI is unlikely to succeed, resulting in wasted resources and increased cybersecurity risks for organizations. By integrating our propositions as design guidelines, developers can create AI systems that align more effectively with the dynamic task-specific requirements of analysts during incident detection and response. Customizing AI systems to accommodate analysts' workflows and cognitive processes would not only facilitate the adoption of AI-powered CSIR tools but also enhance their usability and performance. This, in turn, could improve the overall effectiveness of CSIR and strengthen the cybersecurity posture of organizations by fully leveraging the advantages of AI technology.

6.3 Limitation and Future Work

The mixed method allows us to capture the complex phenomena of analysts' decision-making during an investigation: the interviews reveal the primary concerns and expectations of analysts, while the case study offers insights into non-verbal cues and interactions with AI explanations. However, the approach also comes with several limitations. First, the inherently qualitative nature of the interview study may introduce researchers' bias in the interpretation of the interview data. Furthermore, the professionals we interviewed had only used generative AI in their work setting and lacked experience with learning-based AI models and their explanations, which led us to examine their perceptions rather than their actual experience regarding the potential impact. In Study 2, although we were able to replicate a real-world investigation scenario and integrate AI and explanations, it was infeasible to simulate other aspects of daily routines, such as team interactions, client calls, emails, and similar potential interruptions of analysts' workflow. Moreover, the time commitment necessary to complete the Study 2 tasks and the requirement to be on-site for the experiment led to recruitment challenges, leading to a relatively smaller sample size, limiting the generalizability of our findings.

A primary avenue for future research involves evaluating the efficacy of the proposed interventions in mitigating cognitive load. We anticipate that our proposals may generate several testable hypotheses addressing various facets of the central thesis. For example, studies can examine the optimal delivery method for AI explanations. These approaches may encompass variations between textual and graphic-based information, as well as detailed versus summarized content. Additionally, it would be valuable to explore the effects of detection systems tailored to analysts' individual requirements, considering factors such as cognitive load and situation awareness, on their operational performance. By integrating quantitative analyses with our qualitative findings, future studies would gain a deeper understanding of the analysts' preferences for AI explanations in their task processes and their impact on their performance.

7 Conclusion

In recent years, incident detection and response have garnered significant importance in corporate cybersecurity strategy. Cybersecurity analysts play a vital role in CSIR by analyzing, detecting, and responding to various cyber threats, from malware to phishing attacks and data breaches. Analysts' main tasks involve responding to machine-generated alerts through the identification and investigation of incident indicators. Our study examines the evolving landscape of the cybersecurity analyst's investigation process with the evolving data challenges and the introduction of AI and XAI technologies. Our findings have highlighted the significance of adapting to the analysts' changing needs for AI and AI explanations within the CSIR process. Moving forward, further research could delve deeper into the practical implications of XAI design in the CSIR process, as well as its tangible impact on the job performance of analysts.

Appendix 1: Interview questions

1. Can you describe the role of data in incident detection and investigation and how it has changed over the recent years in terms of available data? What challenges have you faced in terms of the amount of data and how have you addressed them?
2. Could you detail your thought process when you first approach an incident detection and investigation? How does this process evolve as you delve deeper into the analysis?
3. During an incident investigation, how do you organize and structure your analytical process to build a comprehensive understanding of the incident? Can you walk me through your typical thought process when

conducting an incident detection and investigation task?

4. How do you perceive the role of automation in managing and analyzing large volumes of data? Can you provide an example where automation significantly impacted your work?
5. How do you define a “hard problem” in your work? When you face a hard problem, how does your investigation approach change? Can you provide an example?
6. Do you take different cognitive approaches (thought process/analytical strategy) to data analysis when faced with easy versus hard problems?
7. Do you maintain a state of “flow” during your analysis, especially when faced with hard problems?
8. What are the common cognitive challenges you encounter during your investigations? (for example, hard to concentrate, disruptions, lack of deep thinking opportunities, lack of clues for next step, misunderstanding of the tasks)
9. Cybersecurity often involves high-pressure situations and large information volumes. How do you manage cognitive overload during complex investigations?
10. Are there specific factors or situations that tend to disrupt your thought process? How do you manage these disturbances?
11. when you get an explanation of an alert, do you think it may disrupt you?
12. Have you utilized AI-driven tools for incident investigation?
13. If you have used AI tools, how did these tools enhance your ability to detect and respond to threats? “influence your cognitive load during threat detection and response?”
14. If you haven't worked directly with AI tools in cybersecurity, how would you envision their impact on cyber analysis and incident investigation?
15. In your opinion, what are the advantages and limitations of using AI in cybersecurity, especially in the context of data analysis?
16. If given a choice, will you choose to use Gen AI (Chat-GPT) to help with any task? Why or why not?

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Data Availability The de-identified interview transcripts, survey data and experimental data collected during Study 2 will be made available upon request.

Declarations

Ethics Approval and Consent to Participate We obtained IRB approval for both studies.

Consent for Publication Both authors have read and agreed to the published version of the manuscript.

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